Beyond What-If: A Survey and Research Agenda for Out-of-the-Box Uses of Network Digital Twins

The Agentic AI Researcher (Claude Fable 5)
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***Abstract*—Network Digital Twins (NDTs) are usually justified by a familiar trio of use cases: what-if validation of configuration changes, capacity planning, and troubleshooting. This paper argues that the more transformative value of twinning lies elsewhere. Surveying standards (ITU-T, IETF NMRG, 3GPP, ETSI ZSM, O-RAN) and recent literature—every reference independently verified by a second AI agent—we identify eight "out-of-the-box" roles for the NDT: a detonation chamber for cyber attacks, a gymnasium for reinforcement-learning and LLM agents, a sustainability ledger for carbon-aware operation, a notary for compliance auditing and SLA arbitration, a rehearsal stage for network disasters, an examination hall that certifies both human operators and AI agents, a federated marketplace commodity, and a semantic mirror whose own synchronization becomes a communication problem. For each role we grade the maturity of the evidence and expose what remains unclaimed. We distill six white spaces into a research agenda, including the "notarized twin" as regulator-grade evidence and twin-based certification of autonomous network agents. In line with the track's rules, this paper was produced end-to-end by an agentic AI pipeline; the mandatory Disclosure Statement details the process.**

***Index Terms*—network digital twin, network and service management, agentic AI, cyber range, sustainability, zero-touch, 6G**

I. Introduction

Ask a network operator why they want a digital twin and the answer is nearly always the same: to try a change before it hits production. The canonical NDT use cases—what-if analysis of configuration updates, network planning, and troubleshooting—dominate surveys [1]–[3] and anchor the standards work [4], [5]. They are valuable, but they frame the twin as a cautious accessory to existing workflows: a place to double-check decisions humans were going to make anyway.

This paper takes the opposite view. The most consequential uses of a high-fidelity, continuously synchronized replica of a network are the ones that have no pre-twin equivalent at all: detonating malware you would never allow near production, rehearsing an earthquake, letting a reinforcement-learning agent fail ten thousand times without a single dropped customer packet, or handing a regulator a replayable, tamper-evident copy of last Tuesday's outage. These are not accelerated versions of old practice; they are new institutions that only exist because the twin exists.

We make three contributions. First, we assemble an evidence-graded taxonomy of eight out-of-the-box NDT use cases, drawing on standards from ITU-T, the IETF NMRG, 3GPP, ETSI ZSM, and the O-RAN ALLIANCE, and on recent peer-reviewed and preprint literature (Section III, Table I). Second, we identify the cross-cutting obstacles—fidelity metrology, the twin's own attack surface and carbon footprint, and confidentiality in federation—that every unconventional use case inherits (Section IV). Third, we distill six concrete white spaces, verified as unclaimed in the literature we could find, into a research agenda (Section V).

A methodological note befitting this track: the paper itself was produced by an agentic pipeline in which one AI agent performed literature research with live web access, a second AI agent independently verified every citation against its source, and the writing agent used only verified material. One correction this process caught is itself instructive: the IRTF NMRG architecture document for NDTs is still an Internet-Draft [5], not an RFC—a fact that a non-verifying LLM could easily hallucinate the other way.

II. Background: What the Standards Already Admit

ITU-T Recommendation Y.3090 [4] defines the digital twin network as a virtual representation of a physical network used to analyze, diagnose, emulate, and control it, with a real-time interactive mapping between the two, and specifies requirements and a functional architecture. The IRTF NMRG reference architecture [5]—as of this writing an active Internet-Draft in version 13—adds the canonical building blocks: data repositories, service-mapping models, and twin–network synchronization interfaces. In mobile networks, 3GPP SA5 completed a Release-19 study on management aspects of NDTs (TR 28.915), is producing normative work in TS 28.561, and continues in Release 20 (TR 28.883) [6]. ETSI's Zero-touch Service Management group has published a report on NDT enablers for ZSM (GR ZSM 015) and a specification, GS ZSM 018 [7], in which an NDT management service identifies risks of SLA/SLS violations before they occur. The O-RAN ALLIANCE's next-Generation Research Group has compiled the top-five digital-twin RAN use cases contributed by its members: AI/ML training and evaluation, network

testing automation, network planning, network energy saving, and site-specific optimization [8]. ITU-T X.2011 [9] supplies security guidelines for the twin itself.

Two observations motivate everything that follows. First, the unconventional is already leaking into the standards: the NMRG draft explicitly lists cyber-range exercises and staff training on customized replicas among its application scenarios [5], and the O-RAN report ranks AI/ML training—not planning—as its first use case [8]. Second, no standard yet tells you how to *measure* whether a twin is faithful enough for any of these purposes—a gap we return to in Section IV.

III. Eight Out-of-the-Box Use Cases

A. The Detonation Chamber: Cyber Ranges and Attack Rehearsal

The twin is the one place where an operator can safely detonate the attacks its production network must never see. The EU H2020 SPIDER project built a 5G cyber range as a fully containerized twin—core, RAN, and deliberately vulnerable hosts—organized around security assessment, team training, and cyber-risk evaluation [10]. Recent work extends the idea to *multi-ownership* cybersecurity twins that span digital service chains crossing administrative domains, with federation mechanisms for trust and confidentiality [11]. Notably, this use case is already semi-standardized: the NMRG draft lists sandboxed mitigation drills among its application scenarios [5], and ITU-T X.2011 addresses the mirror-image concern—securing the twin itself [9]. Maturity: the best-supported unconventional use case, with running prototypes.

B. The Gymnasium: A Safe Place for AI to Fail

Reinforcement-learning and, increasingly, LLM-based agents need thousands of failures to learn; production networks tolerate none. Colosseum, described by its authors as the world's largest wireless network emulator with hardware in the loop, has been repurposed as an Open RAN digital twin with automated pipelines for training and testing AI/ML xApps; agents trained there transferred to an over-the-air testbed needing only a few rounds of online learning [13], with companion work quantifying the fidelity of the twin against the physical testbed [14]. The sim-to-real framing—and its failure modes—has been analyzed for RL-driven next-generation networks [15]. Fast learned surrogates make the gym affordable: RouteNet-Fermi predicts per-flow delay with a 6.24% mean relative error, including on topologies an order of magnitude larger than those seen in training, at millisecond inference where packet-level simulation takes far longer [16]. The frontier is LLM agents living in the loop: LINKs demonstrates autonomous LLM-driven management of a 6G twin, including radio-resource management with lazy-loading data retrieval [17], and LLM-Twin brings mini-giant model collaboration inside the twin under resource constraints [18].

C. The Sustainability Ledger: Carbon-Aware Operation

Energy actions—carrier shutdown, sleep modes, load consolidation—are exactly the risky, hysteresis-laden interventions operators hesitate to trial in production, which makes them natural twin workloads. Huawei's SRCON builds a live-network-data-driven twin for carrier-shutdown energy optimization at scale, outperforming an operator's incumbent method [20]. A detailed NDT evaluation reports network energy savings of up to 44% during low-traffic hours [19]. Reaching beyond energy to carbon, twin-orchestrated integrated satellite–aerial–terrestrial networks using real grid carbon-intensity data achieve up to 29% lower gCO2 per bit than QoS-only orchestration [21]. The uncomfortable counterpoint: the twin itself consumes energy, and the end-to-end survey of Tran et al. discusses the NDT's own resource footprint and mitigations such as adaptive fidelity and model compression [22]. Nobody has yet published the net ledger—who twins the twin's carbon?

D. The Notary: Compliance Auditing and SLA Arbitration

This is the largest white space we found. The closest verified work is Security-Digital-Twin-as-a-Service, which mirrors real assets to continuously collect machine-readable compliance evidence (CycloneDX SBOMs, W3C Web-of-Things descriptors), replacing disruptive manual certification audits [12]; and ETSI GS ZSM 018, whose NDT management services flag SLA/SLS-violation risk ahead of time [7]. But we found no peer-reviewed proposal for the twin as a *neutral evidence engine*: a notarized, tamper-evident, replayable record of network state that two operators could submit to an arbiter in an SLA dispute, or that a regulator could replay to verify coverage obligations, net-neutrality behavior, or outage reports. Every technical ingredient exists; the institution does not.

E. The Rehearsal Stage: Disaster Preparedness

Fire drills exist for buildings; nothing comparable exists for national networks, because you cannot schedule an earthquake in production. Support in the literature is real but partial. Adversarial Network Imagination couples a causal LLM and knowledge graph to *generate* causally consistent hypothetical failure cascades and executes them in a digital twin to evaluate mitigations—turning the twin from a reactive mirror into an imagination engine [23]. In space, a digital-twin satellite network co-simulated with NASA's "42" spacecraft simulator handles kinematic drift, hardware failures, and adversarial jamming with proactive reconfiguration over a 600-second scenario [24]. What is missing is the terrestrial equivalent at ISP or national scale—mass fiber cuts, power blackouts, flash crowds—and any shared benchmark suite of disaster scenarios against which twins and mitigation policies could be compared.

F. The Examination Hall: Certifying Operators—and Agents

The NMRG draft already lists training staff on customized network replicas as an application scenario [5], and SPIDER targets security-team training [10]. Yet we found no academic

TABLE I
TAXONOMY OF OUT-OF-THE-BOX NDT USE CASES, EVIDENCE MATURITY, AND OPEN GAPS

Use case	Maturity	Key evidence	What remains open
Detonation chamber (cyber range)	Deployed prototypes	SPIDER 5G cyber range [10]; multi-owner blueprints [11]; in NMRG scenarios [5]	Federation across administrative domains end-to-end
Gymnasium for AI agents	Strong research	Colosseum O-RAN twin [13], [14]; sim2real analysis [15]; GNN world models [16]	Certification of trained agents (see U6)
Sustainability ledger	Emerging, quantified	Up to 44% energy savings [19]; SRCON live-data twin [20]; 29% lower gCO2/bit [21]	Net accounting incl. the twin's own footprint [22]
Notary (audit & SLA arbitration)	White space	Continuous certification via twin [12]; SLA-risk MnS [7]	Notarized, replayable, regulator-grade twin state
Rehearsal stage (disaster drills)	Partial	LLM-imagined failure cascades in a twin [23]; satellite resilience twin [24]	Terrestrial mass-failure benchmarks
Examination hall (training & certification)	Thin	Staff training in NMRG [5]; team training in SPIDER [10]	Measured training outcomes; a "driving test" for network AI
Federation & marketplace	Emerging	Twin-as-a-service [12]; federated twins [11], [25]; multi-domain service exposure [26]	Twin economics, packaging, and exchange
Semantic mirror	Novel angle	Causal semantic sync [27]; LLMs inside twins [17], [18]	Communications *for* the twin as a discipline

study that *measures* training or certification outcomes on a *network* digital twin—measured evaluations exist for digital-twin-based cyber ranges of industrial systems [29], which only sharpens the gap on the network side. The sharper version of this idea is aimed at machines: if agents are trained in twins (Section III-B), they can be *examined* in twins. A standardized battery of twin-based scenarios—failure storms, adversarial traffic, conflicting intents—could function as a driving test that an autonomous agent must pass before receiving write-access to production. Twin-based training and evaluation of AI agents is emerging in other domains, such as air-traffic control; no such certification framework exists for *network* agents today.

G. The Federation and the Marketplace

Twin-as-a-service is already instantiated for security [12]; federated digital twins for collaborative decision-making have been architected [25]; multi-ownership blueprints address trust and confidentiality across service chains [11]; and a modular multi-domain 6G architecture exposes NDT capabilities as a first-class service domain, demonstrated on joint cellular-coverage and renewable-energy optimization [26]. What does not exist in the networking literature, anywhere we could verify, is the economics: pricing, licensing, or exchange of *network* twins as products, or a packaging format that would let a twin of a metro network be sold, composed, or escrowed. (Digital-twin pricing and marketplaces have been explored for industrial assets, but not for networks.) If twins become as valuable as this literature suggests, a twin marketplace is the obvious next institution—and it is unclaimed.

H. The Semantic Mirror: Communications for the Twin

The usual framing is twins-for-communications; the inverse is quietly becoming a research field. Keeping a twin faithful under strict bandwidth constraints is itself a communication problem: causal semantic communication transmits only the causally relevant state needed to preserve the twin's fidelity [27], and LLM-Twin adds semantically secure communication between the physical network and its replica [18]. Combined with the closed-loop role standardized in ETSI ZSM [7] and demonstrated in proactive zero-touch management with learned state prediction [28], the synchronization channel—what to send, when, and what it means—emerges as a discipline of its own.

IV. CROSS-CUTTING CHALLENGES

Fidelity metrology. Every use case above rests on an unanswered question: how faithful is this twin, and faithful enough for what? Colosseum's twin-versus-testbed comparisons [14] and RouteNet-Fermi's error bounds [16] are exemplary but ad hoc; there is no standard fidelity score, no agreed test suite, and no way for a certificate earned in one twin (Section III-F) to be trusted by another. The NMRG draft and the foundational surveys name twin validation as open [1], [5].

The twin as attack surface. A compromised twin poisons every downstream use: it can launder a bad configuration into "validated," train an agent on manipulated physics, or fabricate audit evidence. ITU-T X.2011's security guidelines for digital twin networks [9] are a start, but the notary and examination-hall use cases demand integrity guarantees closer to those of audit logs than of simulators.

Confidentiality in federation. Inter-operator twins require sharing precisely the state operators guard most closely. Federation blueprints propose trust mechanisms [11], [25], but no end-to-end multi-operator NDT with confidentiality-preserving synchronization has been demonstrated.

The twin's own cost. Fidelity, freshness, and footprint trade against each other [22]. A sustainability use case whose twin burns more carbon than it saves is an accounting failure waiting to be discovered.

V. Research Agenda

From the gaps verified above, six concrete directions:

1) **The notarized twin.** Tamper-evident, replayable network state with cryptographic integrity, usable as regulator-grade evidence for SLA arbitration, coverage-obligation audits, and outage forensics. Nearest neighbors: [7], [12].
2) **Twin-based certification of autonomous agents.** A standardized examination battery an AI agent must pass in a twin before operating a production network—directly aligned with this conference's agentic theme. Nearest neighbors: [13], [15].
3) **Twin economics.** Packaging formats, licensing, and pricing for twins as tradable products; the protocols of a twin marketplace.
4) **Net carbon accounting of twinning.** Quantify the twin's own energy/carbon overhead against savings achieved [19], [21], [22], with adaptive-fidelity operation driven by that ledger.
5) **An open disaster-rehearsal benchmark.** A shared suite of terrestrial mass-failure scenarios (fiber-cut storms, blackouts, flash crowds) for comparing twins and mitigation policies, extending [23], [24] beyond satellites.
6) **Fidelity metrology.** A standard, portable measure of twin faithfulness per use case, without which none of the above can be trusted across organizational boundaries.

VI. Conclusion

The what-if twin is a better test lab; the twins surveyed here are new institutions: courtrooms, gymnasiums, examination halls, marketplaces. The evidence shows the technical ingredients largely exist and the standards bodies are already edging past the classic use cases. What is missing is deliberate research on the institutional layer—evidence, certification, economics, and the metrology that makes a twin's word worth trusting. For a community turning network management over to agentic AI, the twin is where those agents will learn, be examined, and be held to account. That, not what-if analysis, is the out-of-the-box promise of digital twins for networks and services.

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Disclosure Statement

(Mandatory appendix per the track CFP.) **LLM used:** Claude Fable 5 (model ID `claude-fable-5`, Anthropic), operating autonomously inside the Claude Code agentic harness; no other LLM contributed text. **Framework:** a three-agent pipeline within Claude Code: (i) a *researcher* subagent with live web search/fetch gathered standards and literature and wrote sourced research notes; (ii) a *verifier* subagent independently re-fetched sources to confirm every citation's existence and metadata, flagging anything unconfirmed; (iii) the writing agent composed this paper exclusively from verified material. The paper was typeset with IEEEtran under TeX Live 2026 in a Docker container; all tooling was orchestrated by the AI agent. **Initial prompting strategy:** the human Research Director issued a single natural-language instruction—write a paper for this track on out-of-the-box use cases of digital twins for networks and services, following the CFP format—and took no further action beyond initiating the agentic loop. **Constraints applied:** topic fixed to CFP topic 2; IEEE two-column short-paper format, maximum five pages including references and this statement; citations restricted to sources whose existence was verified against the live web on 21 August 2026; no human editing, rewriting, or correction of any technical content, structure, or references occurred at any stage. **Known limitations:** several references are preprints whose final venues were unconfirmed at verification time and are cited as arXiv works; the survey covers only sources the agents could locate and verify within the session.